

Creating Arrays

```
#concevnction
import numpy as np
#creating 1d array
n1 = np.array([3,4,5,6])
print(n1)

[3 4 5 6]

#creating 2d array
n2 = np.array([[3,5,6,7],[4,5,6,7]])
print(n2)

[[3 5 6 7]
 [4 5 6 7]]

#zero filled 2d array
n3 = np.zeros([3,5],dtype='int32')
print(n3)

[[0 0 0 0 0]
 [0 0 0 0 0]
 [0 0 0 0 0]]

#array filled with ones
n4 = np.ones([2,3,3],dtype='int32')
print(n4)

[[[1 1 1]
   [1 1 1]
   [1 1 1]]
 [[1 1 1]
   [1 1 1]
   [1 1 1]]]

#empty array - contain random junk values
n5 = np.empty([1,3])
print(n5)

[[1.27702224e+232 4.95270032e+223 1.24805535e+243]]

#create array with range of values
n6 = np.arange(10)
print(n6)

[0 1 2 3 4 5 6 7 8 9]
```

```

n7 = np.arange(-5,30,5) #start,stop,step
print(n7)

[-5  0  5 10 15 20 25]

#linearly spaced entries
n8 = np.linspace(0,20,num=8)
print(n8)

[ 0.          2.85714286  5.71428571  8.57142857 11.42857143
14.28571429
17.14285714 20.          ]

n9 = np.full([2,5],99) #create by specifying dimension and fill with values
print(n9)

[[99 99 99 99 99]
 [99 99 99 99 99]]

n5.fill(9) #fill an array with values
print(n5)

[[9. 9. 9.]]

```

Accessing

```

print(n1[0])

3

print(n1[-1])

6

print(n2[1,0])

4

print(n2[1,:])

[4 5 6 7]

print(n2[:,3])

[7 7]

n4[1,1,2] = 10
print(n4)

```

```
[[[ 1  1  1]
   [ 1  1  1]
   [ 1  1  1]]

 [[ 1  1  1]
   [ 1  1 10]
   [ 1  1  1]]]
```

Size and Shape

```
print(n4.ndim) #ndim gives number of dimensions or axes
3
print(n4.shape)
(2, 3, 3)
print(n4.size)
18
print(n4.dtype)
int32
```

Reshaping and Array Conversions

```
n12 = n10.reshape(2,4)
print(n12)

[[ 3  4  1  9]
 [ 5 10  8  7]]

#Joining 2 arrays using concatenate
n13 = np.arange(4)
print(n13)
n14 = np.arange(5,9)
print(n14)
n15 = np.concatenate((n13,n14))
print(n15)

[0 1 2 3]
[5 6 7 8]
[0 1 2 3 5 6 7 8]
```

Indexing and slicing

```
print(n21[3:7])
```

```
[ 3  4  5 100]
```

Math

```
print(n22)
print(n23)
print(n22+n23)
print(n23*n23)
```

```
[[1 2]
 [3 4]]
[[5 6]
 [7 8]]
[[ 6  8]
 [10 12]]
[[25 36]
 [49 64]]
```

```
print(n26)
print(n26.sum())
print(n26.max())
print(n26.min())
print(n26*10)
```

```
[100 200  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16
 17
 18 19]
489
200
2
[1000 2000  20  30  40  50  60  70  80  90 100 110 120 130
 140 150 160 170 180 190]
```

```
print(np.cos(n26))
```

```
[ 0.86231887  0.48718768 -0.41614684 -0.9899925  -0.65364362
 0.28366219
 0.96017029  0.75390225 -0.14550003 -0.91113026 -0.83907153
 0.0044257
 0.84385396  0.90744678  0.13673722 -0.75968791 -0.95765948 -
 0.27516334
 0.66031671  0.98870462]
```

Random Numbers

```
np.random.rand(3,3)
```

```
array([[0.6756754 , 0.58657492, 0.42845491],  
       [0.48703211, 0.25432168, 0.89789337],  
       [0.65050801, 0.58615413, 0.57901873]])
```

```
np.random.randint(100,size=(4,2))
```

```
array([[65, 82],  
       [89, 38],  
       [39, 56],  
       [98, 76]])
```